

ITUS102: Web Technology Framework

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	-	2	1	3	2
Marks	-	50	-	50	

Outline of the Course:

Sr No	Title of the Unit	Minimum number of hours
1.	Web Architecture Fundamentals	4
2.	Client-Side Scripting Fundamentals	8
3.	Server-Side Processing Fundamentals	6
4.	Web-Data Interaction Fundamentals	4
5.	Web Security Threats and Defense Awareness	4
6.	Modern Web Ecosystem	4
Total hours (Practical Lab)		30

Detailed Syllabus:

Sr. No.	Topic / Unit Details	Hours	%
1	Web Architecture Fundamentals	4 Hrs.	12%
	Introduction to Web and Internet, Client-Server Architecture, Web Development Life Cycle, Introduction to Web Services and APIs, Basic Concepts of RESTful APIs, Understanding HTTP Request and Response, HTTP Methods and Status Codes, Introduction to Monolithic and Microservice Architectures, Basics of Session Management, Stateful and Stateless Applications, Client-side vs Server-side Data Storage, Responsive Web Design Principles, CSS Cascading and Inheritance Concepts, Introduction to Version Control Systems, Overview of Git and GitHub.		
2	Client-Side Scripting Fundamentals	8 Hrs.	28%
	Introduction to JavaScript and Browser Execution Environment, Understanding Browser Rendering Process, Variables, Data Types, Operators, Conditional Statements, Loops, Functions, Event Handling Basics, Single-threaded Nature of JavaScript, Event Queue and Event Loop Concepts, DOM Introduction and Element Manipulation, Form Validation using JavaScript, Client-side Data Handling, Introduction to Asynchronous Programming,		

	Timers and Callbacks, Dynamic Page Updates, Browser Storage Concepts, Resource Loading and Performance Awareness, Secure Handling of User Input.		
3	Server-Side Processing Fundamentals	6 Hrs.	20%
	Introduction to Server-side Processing, Understanding Request-Response Lifecycle, Dynamic Content Generation Concepts, Execution Models Overview, Data Access Layers, Database Connectivity Concepts, Connection Management Basics, Input Validation and Sanitization, Authentication and Authorization Concepts, Session and Token-based Authentication Overview, Role-based Access Control Fundamentals, Introduction to Concurrent Requests, Resource Management Basics, External Service Integration Concepts, Synchronous and Asynchronous Communication Overview.		
4	Web-Data Interaction Fundamentals	4 Hrs.	13%
	Introduction to Databases and Web Applications, Data Access Interfaces, Request and Response Flow between Web Applications and Databases, Basic Query Processing Concepts, CRUD Operations, Query Optimization Awareness, Data Caching Fundamentals, Data Consistency Basics, Handling Connection Failures, Introduction to Concurrent Data Access, Connection-Oriented and Connection-Less Communication Concepts.		
5	Web Security Threats and Defense Awareness	4 Hrs.	13%
	Introduction to Web Security, Common Web Vulnerabilities, Secure Password Practices, Session Security Fundamentals, Understanding Timeouts and Retries, Input Validation Concepts, Secure Coding Awareness, Defensive Design Principles, Introduction to Web Hacking Techniques and Ethical Considerations, Safe Browsing Practices, Security Best Practices for Web Applications.		
6	Modern Web Ecosystem	4 Hrs.	14%
	Cross-Browser Compatibility Fundamentals, Browser-Specific Issues and Solutions, Browser Developer Tools for Debugging and Inspection, Mobile-Friendly Web Design Concepts, Responsive Testing Techniques, Introduction to Web Scraping Concepts and Ethical Considerations, Overview of Modern Web Frameworks, Understanding Framework Selection Criteria, Scalability and Performance Awareness, Industry Trends in Web Development.		

Course Outcomes (COs):

At the end of the course, the students will be able to

C01	Design structured and semantic web pages using HTML5.
C02	Apply CSS3 techniques to create responsive and visually appealing web layouts.
C03	Implement UI/UX principles and accessibility standards in web design.
C04	Develop responsive interfaces using Bootstrap and Tailwind CSS frameworks.
C05	Perform validation, testing, debugging, and develop a complete industry standard responsive website.
C06	Evaluate web applications using browser developer tools, and select appropriate modern web frameworks considering scalability, performance, and industry trends.

CO-PO Matrix

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012	PS01	PS02
C01	3	2	2	–	3	–	–	–	–	–	–	1	3	2
C02	2	2	3	–	3	–	–	–	–	–	–	1	3	2
C03	2	3	3	–	2	–	–	1	–	2	–	1	3	2
C04	2	2	3	–	3	–	–	–	–	–	–	1	3	3
C05	2	3	3	3	3	–	–	1	1	2	–	2	3	3
C06	2	3	2	3	3	–	–	–	–	–	–	3	2	3

Recommended Study Material:

Text Books

1. *Jon Duckett, HTML and CSS: Design and Build Websites, Wiley India.*
Provides a beginner-friendly introduction to HTML5 and CSS3 with strong emphasis on visual learning, layout design, and practical webpage development.
2. *Jennifer Niederst Robbins, Learning Web Design, O'Reilly Media.*
Covers HTML, CSS, responsive design, and web standards with hands-on examples suitable for foundational learning.
3. *Uttam K. Roy, Web Technologies, Oxford University Press.*
Introduces core web concepts including client-server architecture, HTML, CSS, and internet fundamentals.
4. *David Flanagan, JavaScript: The Definitive Guide, O'Reilly Media.*
Covers JavaScript programming, DOM manipulation, event handling, browser APIs, and client-side scripting concepts essential for interactive web applications.

Reference Books

1. *Steve Krug, Don't Make Me Think: A Common-Sense Approach to Web Usability, New Riders Publishing.*

Provides practical insights into usability, user experience, navigation design, and creating intuitive web interfaces.

2. *Jakob Nielsen, Usability Engineering, Morgan Kaufmann Publishers.*

A foundational reference for UI/UX principles, accessibility, user-centered design, and usability evaluation techniques.

3. *Marijn Haverbeke, Eloquent JavaScript, No Starch Press.*

An excellent beginner-friendly resource for understanding JavaScript fundamentals, programming logic, and browser-based application development.

4. *Douglas Crockford, JavaScript: The Good Parts, O'Reilly Media.*

Focuses on best practices, coding standards, and effective use of JavaScript for professional web development.

5. *Tanya Janca, Alice and Bob Learn Application Security, Wiley.*

Introduces web security concepts, common vulnerabilities, secure coding practices, and security awareness for modern web applications.

6. *Kyle Simpson, You Don't Know JS Yet (Book Series), O'Reilly Media.*

Provides deeper understanding of JavaScript fundamentals, scope, closures, asynchronous programming, and language internals.